$\sigma^2 > 0$." For any a, $-\infty < a < \infty$, the random variables $Y_1, \ldots, Y_n$ defined by

$$(Y_1, \ldots, Y_n) = g_a(X_1, \ldots, X_n) = (X_1 + a, \ldots, X_n + a)$$

are iid $\mathrm{n}(\mu + a, \sigma^2)$ random variables. Thus, the joint distribution of $\mathbf{Y} = g_a(\mathbf{X})$ is in $\mathcal{F}$ and hence $\mathcal{F}$ is invariant under $\mathcal{G}$. In terms of the notation in Definition 6.4.4, if $\theta = (\mu, \sigma^2)$, then $\theta' = (\mu + a, \sigma^2)$. ||

Remember, once again, that the Equivariance Principle is composed of two distinct types of equivariance. One type, measurement equivariance, is intuitively reasonable. When many people think of the Equivariance Principle, they think that it refers only to measurement equivariance. If this were the case, the Equivariance Principle would probably be universally accepted. But the other principle, formal invariance, is quite different. It equates any two problems with the same mathematical structure, regardless of the physical reality they are trying to explain. It says that one inference procedure is appropriate *even if the physical realities are quite different*, an assumption that is sometimes difficult to justify.

But like the Sufficiency Principle and the Likelihood Principle, the Equivariance Principle is a data reduction technique that restricts inference by prescribing what other inferences must be made at related sample points. All three principles prescribe relationships between inferences at different sample points, restricting the set of allowable inferences and, in this way, simplifying the analysis of the problem.

6.5 Exercises

6.1 Let X be one observation from a $\mathrm{n}(0, \sigma^2)$ population. Is $|X|$ a sufficient statistic?

6.2 Let $X_1, \ldots, X_n$ be independent random variables with densities

$$f_{X_i}(x|\theta) = \begin{cases} e^{i\theta - x} & x \geq i\theta \\ 0 & x < i\theta. \end{cases}$$

Prove that $T = \min_i(X_i/i)$ is a sufficient statistic for θ.

6.3 Let $X_1, \ldots, X_n$ be a random sample from the pdf

$$f(x|\mu, \sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma}, \ \mu < x < \infty, \ 0 < \sigma < \infty.$$

Find a two-dimensional sufficient statistic for (μ, σ).

6.4 Prove Theorem 6.2.10.

6.5 Let $X_1, \ldots, X_n$ be independent random variables with pdfs

$$f(x_i|\theta) = \begin{cases} \frac{1}{2i\theta} & -i(\theta - 1) < x_i < i(\theta + 1) \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta > 0$. Find a two-dimensional sufficient statistic for θ.

6.6 Let $X_1, \ldots, X_n$ be a random sample from a $\mathrm{gamma}(\alpha, \beta)$ population. Find a two-dimensional sufficient statistic for (α, β).

6.7 Let $f(x, y|\theta_1, \theta_2, \theta_3, \theta_4)$ be the bivariate pdf for the uniform distribution on the rectangle with lower left corner (θ_1, θ_2) and upper right corner (θ_3, θ_4) in $\Re^2$. The parameters satisfy $\theta_1 < \theta_3$ and $\theta_2 < \theta_4$. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be a random sample from this pdf. Find a four-dimensional sufficient statistic for $(\theta_1, \theta_2, \theta_3, \theta_4)$.

6.8 Let $X_1, \ldots, X_n$ be a random sample from a population with location pdf $f(x-\theta)$. Show that the order statistics, $T(X_1, \ldots, X_n) = (X_{(1)}, \ldots, X_{(n)})$, are a sufficient statistic for θ and no further reduction is possible.

6.9 For each of the following distributions let $X_1, \ldots, X_n$ be a random sample. Find a minimal sufficient statistic for θ.

(a) $f(x|\theta) = \frac{1}{\sqrt{2\pi}} e^{-(x-\theta)^2/2}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty$ (normal)

(b) $f(x|\theta) = e^{-(x-\theta)}, \quad \theta < x < \infty, \quad -\infty < \theta < \infty$ (location exponential)

(c) $f(x|\theta) = \frac{e^{-(x-\theta)}}{\left(1+e^{-(x-\theta)}\right)^2}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty$ (logistic)

(d) $f(x|\theta) = \frac{1}{\pi[1+(x-\theta)^2]}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty$ (Cauchy)

(e) $f(x|\theta) = \frac{1}{2} e^{-|x-\theta|}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty$ (double exponential)

6.10 Show that the minimal sufficient statistic for the uniform$(\theta, \theta+1)$, found in Example 6.2.15, is not complete.

6.11 Refer to the pdfs given in Exercise 6.9. For each, let $X_{(1)} < \cdots < X_{(n)}$ be the ordered sample, and define $Y_i = X_{(n)} - X_{(i)}, i = 1, \ldots, n-1$.

(a) For each of the pdfs in Exercise 6.9, verify that the set $(Y_1, \ldots, Y_{n-1})$ is ancillary for θ. Try to prove a general theorem, like Example 6.2.18, that handles all these families at once.

(b) In each case determine whether the set $(Y_1, \ldots, Y_{n-1})$ is independent of the minimal sufficient statistic.

6.12 A natural ancillary statistic in most problems is the *sample size*. For example, let N be a random variable taking values $1, 2, \ldots$ with known probabilities $p_1, p_2, \ldots$, where $\Sigma p_i = 1$. Having observed $N = n$, perform n Bernoulli trials with success probability θ, getting X successes.

(a) Prove that the pair (X, N) is minimal sufficient and N is ancillary for θ. (Note the similarity to some of the hierarchical models discussed in Section 4.4.)

(b) Prove that the estimator X/N is unbiased for θ and has variance $\theta(1-\theta)\mathrm{E}(1/N)$.

6.13 Suppose X_1 and X_2 are iid observations from the pdf $f(x|\alpha) = \alpha x^{\alpha-1} e^{-x^\alpha}, x > 0, \alpha > 0$. Show that $(\log X_1)/(\log X_2)$ is an ancillary statistic.

6.14 Let $X_1, \ldots, X_n$ be a random sample from a location family. Show that $M - \bar{X}$ is an ancillary statistic, where M is the sample median.

6.15 Let $X_1, \ldots, X_n$ be iid $\mathrm{n}(\theta, a\theta^2)$, where a is a known constant and $\theta > 0$. $(a>0)$

(a) Show that the parameter space does not contain a two-dimensional open set.

(b) Show that the statistic $T = (\overline{X}, S^2)$ is a sufficient statistic for θ, but the family of distributions is not complete.

6.16 A famous example in genetic modeling (Tanner, 1996 or Dempster, Laird, and Rubin 1977) is a genetic linkage multinomial model, where we observe the multinomial vector (x_1, x_2, x_3, x_4) with cell probabilities given by $(\frac{1}{2} + \frac{\theta}{4}, \frac{1}{4}(1-\theta), \frac{1}{4}(1-\theta), \frac{\theta}{4})$.

(a) Show that this is a curved exponential family.

(b) Find a sufficient statistic for θ.

(c) Find a minimal sufficient statistic for θ.

6.17 Let $X_1, \ldots, X_n$ be iid with geometric distribution

$$P_\theta(X = x) = \theta(1-\theta)^{x-1}, \quad x = 1, 2, \ldots, \quad 0 < \theta < 1.$$

Show that ΣX_i is sufficient for θ, and find the family of distributions of ΣX_i. Is the family complete?

6.18 Let $X_1, \ldots, X_n$ be iid Poisson(λ). Show that the family of distributions of ΣX_i is complete. Prove completeness without using Theorem 6.2.25.

6.19 The random variable X takes the values 0, 1, 2 according to one of the following distributions:

	$P(X=0)$	$P(X=1)$	$P(X=2)$	
Distribution 1	p	$3p$	$1-4p$	$0 < p < \frac{1}{4}$
Distribution 2	p	p^2	$1-p-p^2$	$0 < p < \frac{1}{2}$

In each case determine whether the family of distributions of X is complete.

6.20 For each of the following pdfs let $X_1, \ldots, X_n$ be iid observations. Find a complete sufficient statistic, or show that one does not exist.

(a) $f(x|\theta) = \frac{2x}{\theta^2}, \quad 0 < x < \theta, \quad \theta > 0$

(b) $f(x|\theta) = \frac{\theta}{(1+x)^{1+\theta}}, \quad 0 < x < \infty, \quad \theta > 0$

(c) $f(x|\theta) = \frac{(\log\theta)\theta^x}{\theta - 1}, \quad 0 < x < 1, \quad \theta > 1$

(d) $f(x|\theta) = e^{-(x-\theta)}\exp(-e^{-(x-\theta)}), \quad -\infty < x < \infty, \quad -\infty < \theta < \infty$

(e) $f(x|\theta) = \binom{2}{x}\theta^x(1-\theta)^{2-x}, \quad x = 0, 1, 2, \quad 0 \le \theta \le 1$

6.21 Let X be one observation from the pdf

$$f(x|\theta) = \left(\frac{\theta}{2}\right)^{|x|}(1-\theta)^{1-|x|}, \quad x = -1, 0, 1, \quad 0 \le \theta \le 1.$$

(a) Is X a complete sufficient statistic?

(b) Is $|X|$ a complete sufficient statistic?

(c) Does $f(x|\theta)$ belong to the exponential class?

6.22 Let $X_1, \ldots, X_n$ be a random sample from a population with pdf

$$f(x|\theta) = \theta x^{\theta - 1}, \quad 0 < x < 1, \quad \theta > 0.$$

(a) Is ΣX_i sufficient for θ?

(b) Find a complete sufficient statistic for θ.

6.23 Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on the interval $(\theta, 2\theta)$, $\theta > 0$. Find a minimal sufficient statistic for θ. Is the statistic complete?

6.24 Consider the following family of distributions:

$$\mathcal{P} = \{P_\lambda(X = x) : P_\lambda(X = x) = \lambda^x e^{-\lambda}/x!; x = 0, 1, 2, \ldots; \lambda = 0 \text{ or } 1\}.$$

This is a Poisson family with λ restricted to be 0 or 1. Show that the family $\mathcal{P}$ is *not complete*, demonstrating that completeness can be dependent on the range of the parameter. (See Exercises 6.15 and 6.18.)

6.25 We have seen a number of theorems concerning sufficiency and related concepts for exponential families. Theorem 5.2.11 gave the distribution of a statistic whose sufficiency is characterized in Theorem 6.2.10 and completeness in Theorem 6.2.25. But if the family is curved, the open set condition of Theorem 6.2.25 is not satisfied. In such cases, is the sufficient statistic of Theorem 6.2.10 also minimal? By applying Theorem 6.2.13 to $T(\mathbf{x})$ of Theorem 6.2.10, establish the following:

(a) The statistic $(\sum X_i, \sum X_i^2)$ is sufficient, but not minimal sufficient, in the $\mathrm{n}(\mu, \mu)$ family.
(b) The statistic $\sum X_i^2$ is minimal sufficient in the $\mathrm{n}(\mu, \mu)$ family.
(c) The statistic $(\sum X_i, \sum X_i^2)$ is minimal sufficient in the $\mathrm{n}(\mu, \mu^2)$ family.
(d) The statistic $(\sum X_i, \sum X_i^2)$ is minimal sufficient in the $\mathrm{n}(\mu, \sigma^2)$ family.

6.26 Use Theorem 6.6.5 to establish that, given a sample $X_1, \ldots, X_n$, the following statistics are minimal sufficient.

	Statistic	Distribution
(a)	$\bar{X}$	$\mathrm{n}(\theta, 1)$
(b)	$\sum X_i$	$\mathrm{gamma}(\alpha, \beta)$, α known
(c)	$\max X_i$	$\mathrm{uniform}(0, \theta)$
(d)	$X_{(1)}, \ldots, X_{(n)}$	$\mathrm{Cauchy}(\theta, 1)$
(e)	$X_{(1)}, \ldots, X_{(n)}$	$\mathrm{logistic}(\mu, \beta)$

6.27 Let $X_1, \ldots, X_n$ be a random sample from the *inverse Gaussian distribution* with pdf

$$f(x|\mu, \lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} e^{\frac{-\lambda(x-\mu)^2}{2\mu^2 x}}, \quad 0 < x < \infty.$$

(a) Show that the statistics

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad T = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i} - \frac{1}{\bar{X}}}$$

are sufficient and complete.

(b) For $n = 2$, show that $\bar{X}$ has an inverse Gaussian distribution, $n\lambda/T$ has a χ^2_{n-1} distribution, and they are independent. (Schwarz and Samanta 1991 do the general case.)

The inverse Gaussian distribution has many applications, particularly in modeling of lifetimes. See the books by Chikkara and Folks (1989) and Seshadri (1993).

6.28 Prove Theorem 6.6.5. (*Hint*: First establish that the minimal sufficiency of $T(\mathbf{X})$ in the family $\{f_0(\mathbf{x}), \ldots, f_k(\mathbf{x})\}$ follows from Theorem 6.2.13. Then argue that any statistic that is sufficient in $\mathcal{F}$ must be a function of $T(\mathbf{x})$.)

6.29 The concept of minimal sufficiency can be extended beyond parametric families of distributions. Show that if $X_1, \ldots, X_n$ are a random sample from a density f that is unknown, then the order statistics are minimal sufficient.
(*Hint*: Use Theorem 6.6.5, taking the family $\{f_0(\mathbf{x}), \ldots, f_k(\mathbf{x})\}$ to be logistic densities.)

6.30 Let $X_1, \ldots, X_n$ be a random sample from the pdf $f(x|\mu) = e^{-(x-\mu)}$, where $-\infty < \mu < x < \infty$.

(a) Show that $X_{(1)} = \min_i X_i$ is a complete sufficient statistic.
(b) Use Basu's Theorem to show that $X_{(1)}$ and S^2 are independent.

6.31 Boos and Hughes-Oliver (1998) detail a number of instances where application of Basu's Theorem can simplify calculations. Here are a few.

(a) Let $X_1, \ldots, X_n$ be iid $\mathrm{n}(\mu, \sigma^2)$, where σ^2 is known.

(i) Show that $\bar{X}$ is complete sufficient for μ, and S^2 is ancillary. Hence by Basu's Theorem, $\bar{X}$ and S^2 are independent.

(ii) Show that this independence carries over even if σ^2 is unknown, as knowledge of σ^2 has no bearing on the distributions. (Compare this proof to the more involved Theorem 5.3.1(a).)

(b) A *Monte Carlo swindle* is a technique for improving variance estimates. Suppose that $X_1, \ldots, X_n$ are iid $\mathrm{n}(\mu, \sigma^2)$ and that we want to compute the variance of the median, M.

(i) Apply Basu's Theorem to show that $\mathrm{Var}(M) = \mathrm{Var}(M - \bar{X}) + \mathrm{Var}(\bar{X})$; thus we only have to simulate the $\mathrm{Var}(M - \bar{X})$ piece of $\mathrm{Var}(M)$ (since $\mathrm{Var}(\bar{X}) = \sigma^2/n$).

(ii) Show that the swindle estimate is more precise by showing that the variance of M is approximately $2[\mathrm{Var}(M)]^2/(N-1)$ and that of $M - \bar{X}$ is approximately $2[\mathrm{Var}(M - \bar{X})]^2/(N-1)$, where N is the number of Monte Carlo samples.

(c) (i) If X/Y and Y are independent random variables, show that

$$\mathrm{E}\left(\frac{X}{Y}\right)^k = \frac{\mathrm{E}(X^k)}{\mathrm{E}(Y^k)}.$$

(ii) Use this result and Basu's Theorem to show that if $X_1, \ldots, X_n$ are iid $\mathrm{gamma}(\alpha, \beta)$, where α is known, then for $T = \sum_i X_j$

$$\mathrm{E}\left(X_{(i)} \middle| T\right) = \mathrm{E}\left(\frac{X_{(i)}}{T} T \middle| T\right) = T \frac{\mathrm{E}(X_{(i)})}{\mathrm{E}T}.$$

6.32 Prove the Likelihood Principle Corollary. That is, assuming both the Formal Sufficiency Principle and the Conditionality Principle, prove that if $E = (\mathbf{X}, \theta, \{f(\mathbf{x}|\theta)\})$ is an experiment, then $\mathrm{Ev}(E, \mathbf{x})$ should depend on E and $\mathbf{x}$ only through $L(\theta|\mathbf{x})$.

6.33 Fill in the gaps in the proof of Theorem 6.3.6, Birnbaum's Theorem.

(a) Define $g(\mathbf{t}|\theta) = g((j, \mathbf{x}_j)|\theta) = f^*((j, \mathbf{x}_j)|\theta)$ and

$$h(j, \mathbf{x}_j) = \begin{cases} C & \text{if } (j, \mathbf{x}_j) = (2, \mathbf{x}_2^*) \\ 1 & \text{otherwise.} \end{cases}$$

Show that $T(j, \mathbf{x}_j)$ is a sufficient statistic in the E^* experiment by verifying that

$$g(T(j, \mathbf{x}_j)|\theta) h(j, \mathbf{x}_j) = g((j, \mathbf{x}_j)|\theta)(1) = f^*((j, \mathbf{x}_j)|\theta)$$

for all $(j, \mathbf{x}_j)$.

(b) As T is sufficient, show that the Formal Sufficiency Principle implies (6.3.4). Also the Conditionality Principle implies (6.3.5), and hence deduce the Formal Likelihood Principle.

(c) To prove the converse, first let one experiment be the E^* experiment and the other E_j and deduce that $\mathrm{Ev}(E^*, (j, \mathbf{x}_j)) = \mathrm{Ev}(E_j, \mathbf{x}_j)$, the Conditionality Principle. Then, if $T(\mathbf{X})$ is sufficient and $T(\mathbf{x}) = T(\mathbf{y})$, show that the likelihoods are proportional and then use the Formal Likelihood Principle to deduce $\mathrm{Ev}(E, \mathbf{x}) = \mathrm{Ev}(E, \mathbf{y})$, the Formal Sufficiency Principle.

6.34 Consider the model in Exercise 6.12. Show that the Formal Likelihood Principle implies that any conclusions about θ should not depend on the fact that the sample size n was chosen randomly. That is, the likelihood for (n, x), a sample point from Exercise 6.12, is proportional to the likelihood for the sample point x, a sample point from a fixed-sample-size binomial(n, θ) experiment.

6.35 A risky experimental treatment is to be given to at most three patients. The treatment will be given to one patient. If it is a success, then it will be given to a second. If it is a success, it will be given to a third patient. Model the outcomes for the patients as independent Bernoulli(p) random variables. Identify the four sample points in this model and show that, according to the Formal Likelihood Principle, the inference about p should not depend on the fact that the sample size was determined by the data.

6.36 One advantage of using a minimal sufficient statistic is that unbiased estimators will have smaller variance, as the following exercise will show. Suppose that T_1 is sufficient and T_2 is minimal sufficient, U is an unbiased estimator of θ, and define $U_1 = E(U|T_1)$ and $U_2 = E(U|T_2)$.

(a) Show that $U_2 = E(U_1|T_2)$.

(b) Now use the conditional variance formula (Theorem 4.4.7) to show that $\operatorname{Var} U_2 \leq \operatorname{Var} U_1$.

(See Pena and Rohatgi 1994 for more on the relationship between sufficiency and unbiasedness.)

6.37 Joshi and Nabar (1989) examine properties of linear estimators for the parameter in the so-called "Problem of the Nile," where (X, Y) has the joint density

$$f(x, y|\theta) = \exp\{-(\theta x + y/\theta)\}, \qquad x > 0, \quad y > 0.$$

(a) For an iid sample of size n, show that the Fisher information is $I(\theta) = 2n/\theta^2$.

(b) For the estimators

$$T = \sqrt{\sum Y_i / \sum X_i} \quad \text{and} \quad U = \sqrt{\sum X_i \sum Y_i},$$

show that

(i) the information in T alone is $[2n/(2n+1)]I(\theta)$;

(ii) the information in (T, U) is $I(\theta)$;

(iii) (T, U) is jointly sufficient but not complete.

6.38 In Definition 6.4.2, show that (iii) is implied by (i) and (ii).

6.39 Measurement equivariance requires the same inference for two equivalent data points: $\mathbf{x}$, measurements expressed in one scale, and $\mathbf{y}$, *exactly the same measurements* expressed in a different scale. Formal invariance, in the end, leads to a relationship between the inferences at two *different* data points in the same measurement scale. Suppose an experimenter wishes to estimate θ, the mean boiling point of water, based on a single observation X, the boiling point measured in degrees Celsius. Because of the altitude and impurities in the water he decides to use the estimate $T(x) = .5x + .5(100)$. If the measurement scale is changed to degrees Fahrenheit, the experimenter would use $T^*(y) = .5y + .5(212)$ to estimate the mean boiling point expressed in degrees Fahrenheit.

(a) The familiar relation between degrees Celsius and degrees Fahrenheit would lead us to convert Fahrenheit to Celsius using the transformation $\frac{5}{9}(T^*(y) - 32)$. Show that this procedure is measurement equivariant in that the same answer will be obtained for the same data; that is, $\frac{5}{9}(T^*(y) - 32) = T(x)$.

(b) Formal invariance would require that $T(x) = T^*(x)$ for all x. Show that the estimators we have defined above do not satisfy this. So they are not equivariant in the sense of the Equivariance Principle.

6.40 Let $X_1, \ldots, X_n$ be iid observations from a location–scale family. Let $T_1(X_1, \ldots, X_n)$ and $T_2(X_1, \ldots, X_n)$ be two statistics that both satisfy

$$T_i(ax_1 + b, \ldots, ax_n + b) = aT_i(x_1, \ldots, x_n)$$

for all values of $x_1, \ldots, x_n$ and b and for any $a > 0$.

(a) Show that T_1/T_2 is an ancillary statistic.

(b) Let R be the sample range and S be the sample standard deviation. Verify that R and S satisfy the above condition so that R/S is an ancillary statistic.

6.41 Suppose that for the model in Example 6.4.6, the inference to be made is an estimate of the mean μ. Let $T(\mathbf{x})$ be the estimate used if $\mathbf{X} = \mathbf{x}$ is observed. If $g_a(\mathbf{X}) = \mathbf{Y} = \mathbf{y}$ is observed, then let $T^*(\mathbf{y})$ be the estimate of $\mu + a$, the mean of each Y_i. If $\mu + a$ is estimated by $T^*(\mathbf{y})$, then μ would be estimated by $T^*(\mathbf{y}) - a$.

(a) Show that measurement equivariance requires that $T(\mathbf{x}) = T^*(\mathbf{y}) - a$ for all $\mathbf{x} = (x_1, \ldots, x_n)$ and all a.

(b) Show that formal invariance requires that $T(\mathbf{x}) = T^*(\mathbf{x})$ and hence the Equivariance Principle requires that $T(x_1, \ldots, x_n) + a = T(x_1 + a, \ldots, x_n + a)$ for all $(x_1, \ldots, x_n)$ and all a.

(c) If $X_1, \ldots, X_n$ are iid $f(x - \theta)$, show that, as long as $\mathrm{E}_0 X_1 = 0$, the estimator $W(X_1, \ldots, X_n) = \bar{X}$ is equivariant for estimating θ and satisfies $\mathrm{E}_\theta W = \theta$.

6.42 Suppose we have a random sample $X_1, \ldots, X_n$ from $\frac{1}{\sigma} f((x - \theta)/\sigma)$, a location–scale pdf. We want to estimate θ, and we have two groups of transformations under consideration:

$$\mathcal{G}_1 = \{g_{a,c}(\mathbf{x})\colon -\infty < a < \infty,\ c > 0\},$$

where $g_{a,c}(x_1, \ldots, x_n) = (cx_1 + a, \ldots, cx_n + a)$, and

$$\mathcal{G}_2 = \{g_a(\mathbf{x})\colon -\infty < a < \infty\},$$

where $g_a(x_1, \ldots, x_n) = (x_1 + a, \ldots, x_n + a)$.

(a) Show that estimators of the form

$$W(x_1, \ldots, x_n) = \bar{x} + k,$$

where k is a nonzero constant, are equivariant with respect to the group $\mathcal{G}_2$ but are not equivariant with respect to the group $\mathcal{G}_1$.

(b) For each group, under what conditions does an equivariant estimator W satisfy $E_\theta W = \theta$, that is, it is unbiased for estimating θ?

6.43 Again, suppose we have a random sample $X_1, \ldots, X_n$ from $\frac{1}{\sigma} f((x - \theta)/\sigma)$, a location–scale pdf, but we are now interested in estimating σ^2. We can consider three groups of transformations:

$$\mathcal{G}_1 = \{g_{a,c}(\mathbf{x})\colon -\infty < a < \infty,\ c > 0\},$$

where $g_{a,c}(x_1, \ldots, x_n) = (cx_1 + a, \ldots, cx_n + a)$;

$$\mathcal{G}_2 = \{g_a(\mathbf{x}) \colon -\infty < a < \infty\},$$

where $g_a(x_1, \ldots, x_n) = (x_1 + a, \ldots, x_n + a)$; and

$$\mathcal{G}_3 = \{g_c(\mathbf{x}) \colon c > 0\},$$

where $g_c(x_1, \ldots, x_n) = (cx_1, \ldots, cx_n)$.

(a) Show that estimators of σ^2 of the form kS^2, where k is a positive constant and S^2 is the sample variance, are invariant with respect to $\mathcal{G}_2$ and equivariant with respect to the other two groups.

(b) Show that the larger class of estimators of σ^2 of the form

$$W(X_1, \ldots, X_n) = \phi\left(\frac{\bar{X}}{S}\right) S^2,$$

where $\phi(x)$ is a function, are equivariant with respect to $\mathcal{G}_3$ but not with respect to either $\mathcal{G}_1$ or $\mathcal{G}_2$, unless $\phi(x)$ is a constant (Brewster and Zidek 1974). Consideration of estimators of this form led Stein (1964) and Brewster and Zidek (1974) to find improved estimators of variance (see Lehmann and Casella 1998, Section 3.3).

6.6 Miscellanea

6.6.1 *The Converse of Basu's Theorem*

An interesting statistical fact is that the converse of Basu's Theorem is false. That is, if $T(\mathbf{X})$ is independent of every ancillary statistic, it does not necessarily follow that $T(\mathbf{X})$ is a complete, minimal sufficient statistic. A particularly nice treatment of the topic is given by Lehmann (1981). He makes the point that one reason the converse fails is that ancillarity is a property of the *entire distribution* of a statistic, whereas completeness is a property dealing only with *expectations*. Consider the following modification of the definition of ancillarity.

Definition 6.6.1 A statistic $V(\mathbf{X})$ is called *first-order ancillary* if $\mathrm{E}_\theta V(\mathbf{X})$ is independent of θ.

Lehmann then proves the following theorem, which is somewhat of a converse to Basu's Theorem.

Theorem 6.6.2 *Let T be a statistic with* $\operatorname{Var} T < \infty$. *A necessary and sufficient condition for T to be complete is that every bounded first-order ancillary V is uncorrelated (for all θ) with every bounded real-valued function of T.*

Lehmann also notes that a type of converse is also obtainable if, instead of modifying the definition of ancillarity, the definition of completeness is modified.

6.6.2 Confusion About Ancillarity

One of the problems with the concept of *ancillarity* is that there are many different definitions of ancillarity, and different properties are given in these definitions. As was seen in this chapter, ancillarity is confusing enough with one definition—with five or six the situation becomes hopeless.

As told by Buehler (1982), the concept of ancillarity goes back to Sir Ronald Fisher (1925), "who left a characteristic trail of intriguing concepts but no definition." Buehler goes on to tell of at least *three* definitions of ancillarity, crediting, among others, Basu (1959) and Cox and Hinkley (1974). Buehler gives eight properties of ancillary statistics and lists 25 examples.

However, it is worth the effort to understand the difficult topic of ancillarity, as it can play an important role in inference. Brown (1996) shows how ancillarity affects inference in regression, and Reid (1995) reviews the role of ancillarity (and other conditioning) in inference. The review article of Lehmann and Scholz (1992) provides a good entry to the topic.

6.6.3 More on Sufficiency

1. *Sufficiency and Likelihood*

   There is a striking similarity between the statement of Theorem 6.2.13 and the Likelihood Principle. Both relate to the ratio $L(\theta|\mathbf{x})/L(\theta|\mathbf{y})$, one to describe a minimal sufficient statistic and the other to describe the Likelihood Principle. In fact, these theorems can be combined, with a bit of care, into the fact that a statistic $T(\mathbf{x})$ is a minimal sufficient statistic if and only if it is a one-to-one function of $L(\theta|\mathbf{x})$ (where two sample points that satisfy (6.3.1) are said to have the same likelihood function). Example 6.3.3 and Exercise 6.9 illustrate this point.

2. *Sufficiency and Necessity*

   We may ask, "If there are *sufficient* statistics, why aren't there *necessary* statistics?" In fact, there are. According to Dynkin (1951), we have the following definition.

   Definition 6.6.3 A statistic is said to be *necessary* if it can be written as a function of every sufficient statistic.

   If we compare the definition of a necessary statistic and the definition of a minimal sufficient statistic, it should come as no surprise that we have the following theorem.

   Theorem 6.6.4 *A statistic is a minimal sufficient statistic if and only if it is a necessary and sufficient statistic.*

3. *Minimal Sufficiency*

   There is an interesting development of minimal sufficiency that actually follows from Theorem 6.2.13 (see Exercise 6.28) and is extremely useful in establishing minimal sufficiency outside of the exponential family.